

Genetic diversity affects ecosystem functions across trophic levels as much as species diversity, but in an opposite direction

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This **important** study provides empirical evidence of the effects of genetic diversity and species diversity on ecosystem functions across multi-trophic levels in an aquatic ecosystem. The support for these findings is **solid**, but a more nuanced interpretation of the results could strengthen the conclusions. The work will be of interest to ecologists working on multi-trophic relationships and biodiversity.

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Abstract

Understanding the relationships between biodiversity and ecosystem functioning stands as a cornerstone in ecological research. Extensive evidence now underscores the profound impact of species loss on the stability and dynamics of ecosystem functions. However, it remains unclear whether the loss of genetic diversity within key species yield similar consequences. Here, we delve into the intricate relationship between species diversity, genetic diversity, and ecosystem functions across three trophic levels —primary producers, primary consumers, and secondary consumers— in natural aquatic ecosystems. Our investigation involves estimating species diversity and genome-wide diversity -gauged within three pivotal species- within each trophic level, evaluating seven key ecosystem functions, and analyzing the magnitude of the relationships between biodiversity and ecosystem functions (BEFs). We found that, overall, the absolute effect size of genetic diversity on ecosystem functions mirrors that of species diversity in natural ecosystems. We nonetheless unveil a striking dichotomy: while genetic diversity was positively correlated with various ecosystem functions, species diversity displays a negative correlation with these functions. These intriguing antagonist effects of species and genetic diversity persists across the three trophic levels (underscoring its systemic nature), but were apparent only when BEFs were assessed within trophic levels rather than across them. This study reveals the complexity of predicting the consequences of genetic and species diversity loss under natural conditions, and

emphasizes the need for further mechanistic models integrating these two facets of biodiversity.

Introduction

Diversity *within* and *among* species are both important to ensure and stabilize ecosystem functions (Cardinale *et al.* 2012 [↗](#); Raffard *et al.* 2019 [↗](#)). Studies on the links between biodiversity and ecosystem functioning (BEFs) have primarily focused on the interspecific (species) facet of biodiversity (Balvanera *et al.* 2006 [↗](#); Hooper *et al.* 2005 [↗](#)). However, the intraspecific (genetic) facet of biodiversity has also recently been shown to have substantial effects on ecosystem functions (Crutsinger *et al.* 2006 [↗](#); Hughes & Stachowicz 2004 [↗](#); Reusch *et al.* 2005 [↗](#)). Recent meta-analyses have shown that genetic diversity of plant and animal populations affect ecosystem functions, and that the magnitude (and shape) of intraspecific BEFs is similar to that of species diversity (Raffard *et al.* 2019 [↗](#); Wan *et al.* 2022 [↗](#)).

Although natural assemblages encompass both intra- and interspecific diversity, most studies investigating BEFs are considering each biodiversity facet separately (but see, Fridley & Grime 2010 [↗](#); Prieto *et al.* 2015 [↗](#); Grele *et al.* 2024 [↗](#)). This makes it difficult to differentiate the relative role of genetic and species diversity in ecosystem functions, impeding general predictions regarding the consequences of biodiversity loss as a whole on ecosystem functions (Blanchet *et al.* 2023 [↗](#)). For instance, we are currently unaware whether the loss of genetic diversity within a few species in an assemblage is as detrimental for ecosystem functions as a species loss, or whether the combined loss of genetic and species diversity may have non-additive consequences for ecosystem dynamics. Although these biodiversity loss scenarios are realistic, our knowledge on the relative role of genetic vs. species diversity in ecosystem functions are still too scarce to provide reliable predictions.

The few studies investigating the combined effects of genetic and species diversity on ecosystem functions were all conducted experimentally by manipulating the genetic and species diversity of assemblages under controlled conditions (Fridley & Grime 2010 [↗](#); Hargrave *et al.* 2011 [↗](#); Prieto *et al.* 2015 [↗](#); but see Grele *et al.* 2024 [↗](#)). Our understanding of genetic (intraspecific) and species (interspecific) BEFs therefore relies on simplified ecosystems that often lack variation in other factors (including spatial scales, abiotic factors...), and in which feedbacks between ecosystem functions and biodiversity are limited (Duffy *et al.* 2017 [↗](#); Prunier *et al.* 2023 [↗](#)). However, knowledge acquired from BEFs at the interspecific level reveals that environmental variation can either reduce or enhance the effects of biodiversity on ecosystem functions, hence generating large variance in the magnitude and direction of BEFs measured in the wild (Hagan *et al.* 2021 [↗](#); Van Der Plas 2019 [↗](#)). One can therefore predict that, under natural conditions, the relative influence of genetic and species diversity on ecosystem functions may deviate from what has been quantified under controlled conditions, although it is difficult to predict the direction of this deviation as field studies (in particular for genetic BEFs) are too scarce to generate clear predictions. Therefore, we need further realistic field studies of BEFs, embracing the whole diversity of life forms (from genes to species) and across realistic environmental gradients to test whether -under natural conditions-species and genetic BEFs are of similar magnitude.

BEF studies often consider a single trophic level, despite accumulating evidence that biodiversity at a given trophic level can propagate across trophic levels, generating “multi-trophic BEFs” (Lefcheck *et al.* 2015 [↗](#); Soliveres *et al.* 2016 [↗](#), Seibold *et al.* 2018 [↗](#)). In particular, studies testing the joint effects of genetic and species diversity on ecosystem functions have mostly considered the effect of primary producer diversity on their own productivity (“*within-trophic level BEFs*”, e.g., Hargrave *et al.* 2011 [↗](#); Prieto *et al.* 2015 [↗](#)). However, genetic and species diversity within a given trophic level may have propagating effects on the ecosystem at other trophic levels (hereafter, “*between-trophic level BEFs*”). Indeed, it is predicted that a genetically-diverse predator population

shares their resources more efficiently than a genetically-poor predator population, which might permit a higher prey species coexistence and hence a larger prey biomass (*between-trophic level* BEFs due to genetic diversity, e.g., Raffard *et al.* 2021 [↗](#)). Alternatively, a species-rich community of primary producers likely exhibits higher primary production, as organisms in species-rich communities share basal resources more efficiently than in species-poor communities (*within-trophic level* BEFs due to species diversity, Balvanera *et al.* 2006 [↗](#); Hooper *et al.* 2005 [↗](#)). Similarly, the relative impact of genetic and species diversity should be inconsistent across trophic. At higher trophic levels (e.g., predators), species richness is generally lower, which should increase the likelihood for genetic diversity (of a few species) to have strong effects on functions. A simple prediction might therefore be that the relative impact of genetic diversity on ecosystem functions should increase with increasing trophic levels (Blanchet *et al.* 2020 [↗](#)). Studies considering genetic and species BEFs under a realistic multitrophic scenario may thus help understanding the trophic contexts under which either genetic or species diversity is more impactful on ecosystem functions than the other, and to test whether genetic and species effects can propagate across trophic levels or not (Seibold *et al.* 2018 [↗](#), Li *et al.* 2020 [↗](#), Moi *et al.* 2021 [↗](#)).

Here, we conducted a field study to test the relative importance of genetic and species diversity for ecosystem functions across multiple trophic levels in a natural landscape. We focused on three trophic levels from river ecosystems; riparian trees (primary producers), macroinvertebrate shredders (primary consumers) and fish (secondary consumers). For each trophic level, we quantified the species diversity of each community, as well as the genetic diversity of a single target and dominant species (*Alnus glutinosa*, *Gammarus* sp. and *Phoxinus phoxinus* respectively). We further estimated several ecosystem functions, including leaf decomposition of riparian trees, biomass (as productivity estimates) of each target species and total biomass of each community within each trophic level. We relied on causal analyses, taking into account the direct and indirect effects of the environment (through biodiversity) on ecosystem functions (Duffy *et al.* 2016 [↗](#)) to test i) whether BEFs measured at the genetic level (*genetic* BEFs) are similar in magnitude and direction to BEFs measured at the species level (*species* BEFs); and ii) whether *within-trophic level* BEFs are similar in magnitude than *between-trophic level* BEFs. We also tested whether the relative effects of species and genetic diversity on ecosystem functions (within or between trophic levels) are consistent across the three trophic levels (primary producers, primary consumers and secondary consumers), in order to generalize findings along the trophic chain. We predicted that -contrary to what has been observed under controlled conditions- *genetic* BEFs and *species* BEFs will not be similar in magnitude, especially because environmental variation may modulate each of them differentially. We further expected that significant genetic and species BEFs will be observed both *within*- and *between*-trophic levels, leading to *within*- and *between*-trophic levels of similar magnitude. Finally, we predicted that the magnitude of *genetic* BEFs will be higher (than that of *species* BEFs) at the highest trophic level (secondary consumers) than at the lowest trophic level (primary producers), mainly because species richness at higher trophic levels presents a lower gradient than at the lowest trophic levels.

Materials and methods

Sampling sites and trophic chain

We sampled 52 sites in Southern France from the Adour-Garonne watershed, and distributed along an east-west gradient in the Pyrenees Mountains (**Figure 1a** [↗](#)). We acquired data on species diversity, genetic diversity and ecosystem functions at three trophic levels (primary producers, primary consumers and secondary consumers) (**Figure 1b** [↗](#)). Riparian trees (57 species in the sampled area) provide organic matter in the form of fallen leaves as a food source for decomposers. We selected the common alder *Alnus glutinosa* for acquisition of genetic data due to its dominance at most sites and its functional relevance, as its roots serve as shelters for many aquatic species and are involved in nitrogen fixation. Macroinvertebrate shredders (101 genera in

the sampled area) are primary consumers using leaves as resources, and converting them into accessible organic matter for other species. We focused on the most abundant Gammarid (Crustacean) species for genetic data acquisition, referred to as *Gammarus* sp. This species has not yet been formally named although it is phylogenetically distinct from its closest relative, *Gammarus fossarum* (Carnevali 2022 [↗](#); Piscart, unpublished data). This species is particularly efficient at decomposing tree leaves, in particular those from *Alnus* (Macneil *et al.* 1997 [↗](#)). Fish (20 species in the sampled area) are secondary consumers feeding on invertebrates (amongst others). We used the minnow *Phoxinus phoxinus* as the fish target species as it is an abundant and important predator strongly impacting invertebrate communities (Raffard *et al.* 2021 [↗](#)).

Biodiversity estimates

Species datasets

At each site, we collected data on the abundance of all species within each trophic level, at one occasion for trees (July-August 2021) and two occasions for invertebrates (July and November 2020) and fishes (mid-July to mid-August 2020 and 2021), to obtain accurate biodiversity estimates. We identified tree species along a 200 m transect of each river bank, excluding trees with trunk smaller than 2 cm in diameter and more than one meter away from the bank. The abundances of trees were estimated as the total number of individuals per species and per site. For invertebrates, we identified shredders to the genus level (or to the family level for some groups such as chironomids) sampled from two types of standardized traps installed in four micro-habitats distributed along the 200m transect used to identify trees: natural coconut brushes (15*5.5 cm, bristles length 7.5 cm) recovered after 1.5 month of colonization, and litter bags (15*11 cm, 0.8 cm mesh size) filled with senescent *Alnus* leaves from each site and recovered after nine days of colonization (see below). We calculated abundances of each genus by summing the number of individuals per genus found in the coco brushes and the litter bags, and we averaged the abundances over the two sampling occasions to get a single estimate per genus per site. For fish, we collected all specimens during single-pass electric fishing sessions over a mean area of $\sim 469.9 \text{ m}^2$ ($\pm 174 \text{ m}^2$) distributed along the 200 m transect. We anesthetized, identified and counted individuals at the species level. We calculated fish abundances as the number of individuals per species and per m^2 , and we averaged the abundances over the two sampling occasions as for invertebrates. Fish species number varies from 1 to 11, invertebrate genus number varies from 15 to 42 and the tree species number varies from 7 to 20 (see Fargeot *et al.* 2023 [↗](#) for details).

Genetic datasets

At each site, we collected tissue from up to 32 individuals of each of the three target species. We collected fresh leaves of *A. glutinosa* in May 2020, specimens of *Gammarus* sp. in February 2020, and a piece of pelvic fin from *P. dragarum* individuals in summer 2020. The DNA of these samples was extracted using commercial kits for *Alnus* and *Gammarus* sp. and a salt-extraction protocol for *P. dragarum* (see Fargeot *et al.* 2023 [↗](#) for details). For each specimen, DNA concentrations were measured using Qubit 3.0 fluorometer (Life Technologies®, USA). Sequencing was performed based on equimolar pools of DNA (“pool-seq” approach, Schlötterer *et al.* 2014 [↗](#)) from each population and each species. For *Gammarus* sp., we also obtained a ~ 600 bp mitochondrial sequence from the COI mitochondrial gene from each individual to ensure identification and avoid mixing individuals from different species. *Gammarus* sp. was found allopatric in most sites, but for a few sites from the eastern part of the area in which two species were identified (Carnevali 2022 [↗](#)). In this latter case, we conserve only the target species for creating the DNA pools. We created one DNA pool per site per species (52 pools for *A. glutinosa*, 47 pools for *Gammarus* sp. and 44 pools for *P. dragarum*) and performed double-digest restriction-site associated DNA sequencing (ddRAD-seq) for *A. glutinosa* and *Gammarus* sp. (respectively, PstI/MseI and Pst/HindIII enzymes) and normalized Genotyping-by-Sequencing (nGBS) for *P. dragarum* (MsiI enzyme). Library

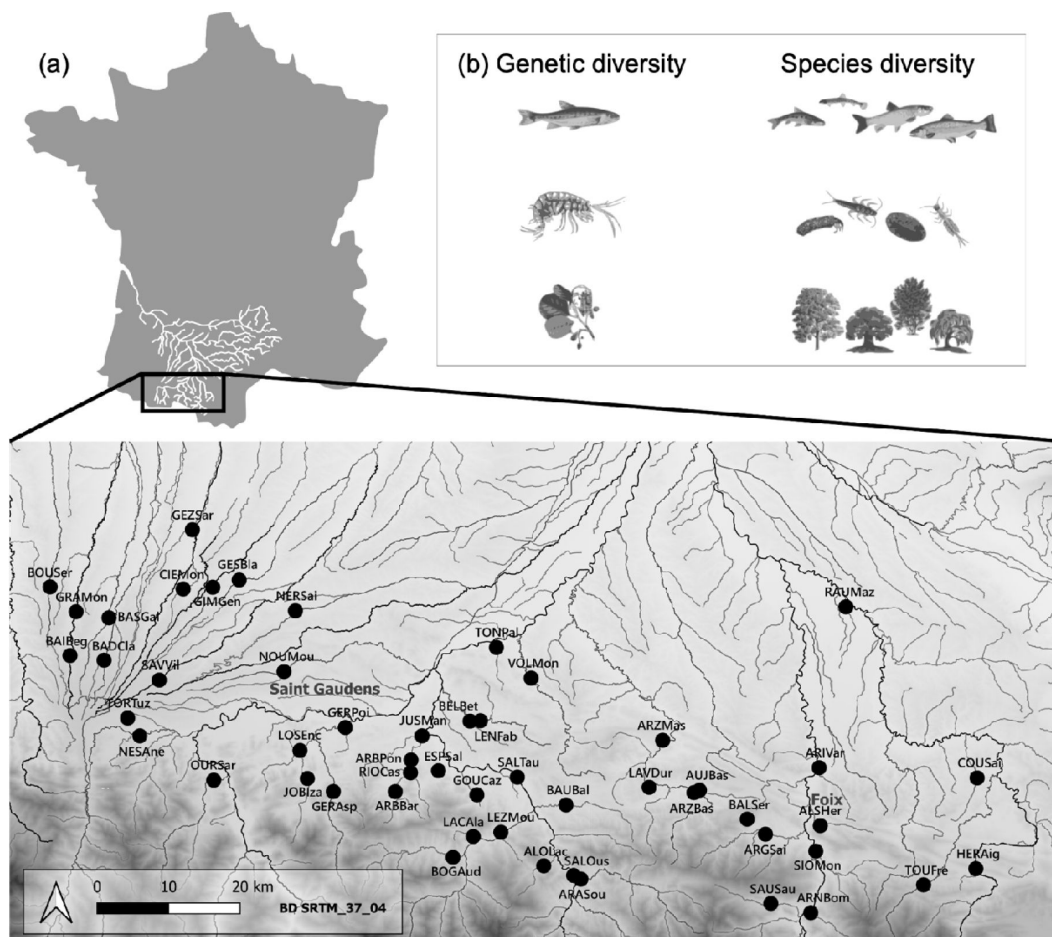


Figure 1.

Location of sampling sites and illustration of the trophic chain.

(a) Distribution of the 52 sampling sites (black dots) spanning an East-West gradient at the foothills of the Pyrenees Mountains (France). Each site is denoted by a six-letter code, with the three first letters indicating the river name and the three last letters indicating the closest city or village. (b) Our study focused on a tri-trophic food chain commonly found in mountain rivers, consisting of riparian trees, macroinvertebrate shredders and fishes (from bottom to top). Within each trophic level, we measured two facets of biodiversity: genetic diversity in a single target species within each trophic level (specifically *Phoxinus phoxinus*, *Gammarus* sp. and *Alnus glutinosa*), and species diversity from communities.

preparation and pool-sequencing were executed by LGC Genomics (Biosearch Technologies®, Germany) on an Illumina NovaSeq® (2×150 pb). Data processing was performed following [De Kort et al. \(2018\)](#), except that read mapping was performed on reference genomes. The genome of *A. glutinosa* was already available ([Griesmann et al. 2018](#)), and we assembled reference genomes from Illumina short-read sequencing and PacBio long-read sequencing for *Gammarus* sp. (available upon request) and *P. dragarum* (accession number on DDBJ/ENA/GenBank: JARPMJ000000000), respectively. SNP calling was performed with (i) filtering of raw sequencing files; (ii) indexing of reference genomes; (iii) mapping reads to the reference; (iv) filtering for unpaired and badly/non-mapped reads; (v) assembling all read information in a single file per population and per species and (vi) calculating SNP allelic frequencies ([De Kort et al. 2018](#)). The total numbers of SNPs retrieved were 583 862 for *A. glutinosa*, 331 728 for *Gammarus* sp. and 414 213 for *P. dragarum* (see [Fargeot et al. 2023](#) for details).

Species and genetic diversity estimates

We calculated LJ-diversity per site using the Shannon entropy from the “hillR” R package for both species and genetic diversity. The Shannon entropy is a metric of evenness that takes into account the distribution of allele or species abundances within each site ([Chao et al. 2014](#)) by weighting each species/allele by its proportional abundance ($q = 1$). Results were similar when using the Simpson’s diversity index ($q = 2$, results not shown).

Ecosystem function measurements

At each site, we measured seven ecosystem functions. We collected biomass production data of all species at each trophic level (hereafter “total biomass”) and the biomass production of each target species as estimates of productivity, as well as the decomposition rate of *Alnus* leaves. Productivity—as we quantified it—is obviously affected by local environmental characteristics, and for this reason, we took into account these potential environmental effects (see hereafter). For riparian tree biomass, we used the trunk diameter of each single tree as a proxy of individual tree biomass, and we summed the trunk diameters of all trees found along the transect (divided by the length of the transect) to estimate the total tree biomass per site and per meter of bank. The same approach was used to estimate *A. glutinosa* biomass. For macroinvertebrate shredders, we estimated the total invertebrate biomass by drying all individuals for 24 h at 60 °C before weighing them (10^{-4} g precision). The same procedure was used to estimate the biomass of *Gammarus* sp. For both estimates, we averaged biomasses over the two sampling sessions. For fish, (fresh) total fish biomass was estimated as the total weight of all individuals (0.01 g precision) per site, whereas *P. dragarum* biomass was the mass of all *P. dragarum* specimens per site. Fish biomasses were averaged over the two sampling sessions.

For the decomposition rate, we quantified leaf mass loss in litter bags placed in four micro-habitats per site twice (July and November 2020). We gathered and dried senescent leaves during fall 2019 from five *Alnus* trees per site to limit individual-specific effects on decomposition. Litter bags were 15 cm x 11 cm pockets of plastic-wire mesh (mesh size; 8 mm to allow invertebrates colonization) in which we introduced 4g of dried leaves before closing the bags with staples. We installed three bags per micro-habitat (12 per site) that we removed sequentially after ~9 days, ~18 days and ~27 days respectively to estimate decomposition rates. Bags were brought back to the laboratory, the remaining leaves were cleaned, dried and weighed. Decomposition rate was estimated as the slope of leaf mass loss over time (obtained from a linear model) that we averaged across replicates and temporal sessions ([Raffard et al. 2021](#)).

Environmental data

A major challenge for inferring BEFs from empirical data is to take into account the direct and indirect (through biodiversity) effects of environmental factors on ecosystem functions ([Duffy et al. 2016](#), [2017](#)). Failing to do this may result in overestimated and/or artefactual BEFs, especially

if the same environmental factor simultaneously affects biodiversity and ecosystem processes (Grace *et al.* 2016 [↗](#)). For each site, we measured thirteen variables related to river topography and physico-chemical characteristics that likely influence biodiversity and ecosystem processes (Altermatt 2013 [↗](#)). *River bed width* (m) was averaged from five measurements per site. *Connectivity* was calculated as the “closeness centrality”, *i.e.*, the inverse of the sum of the distances of a node to all other nodes along the shortest paths possible (Altermatt 2013 [↗](#)), using QGIS and the “RiverDist” R package. *Altitude* (m), *distance from the outlet* (m) and *east-west gradient* (longitudinal position along the Pyrenees chain) were measured using QGIS; *oxygen concentration* (mg.L⁻¹), *oxygen saturation* (%), *water temperature* (°C), *specific conductivity* (µS/cm) and *pH* were measured (and averaged) in summers 2020 and 2021 using a multi-parameter probe (Aqua TROLL 500, In-Situ Inc.). Concentration of NO_3^- , NO_2^- , NH_4^+ and PO_4^{3-} were estimated (and averaged) during summers 2020 and 2021 from a filtered water volume (100 mL) using the Alpkem Flow Solution Iv Autoanalyzer (OI Analytical®).

A Principal Component Analysis combining all thirteen variables was performed using the R package “ade4” (Dray & Dufour 2007 [↗](#)), and coordinates of each site on the two first axes (38.03% of the total variance, see **Table 1** [↗](#)) were used as two synthetic environmental variables for further analyses. We kept only these two first axes to avoid collinearity and over-parameterization of subsequent models. The first axis is defined by a strong contribution of (in decreasing order) oxygen concentration and altitude (**Table 1** [↗](#)). The second axis is defined by a strong contribution of east-west gradient and connectivity (**Table 1** [↗](#)).

Statistical analyses

BEF relationships

To quantify the magnitude of association between biodiversity estimates and ecosystem functions (BEFs), we performed piecewise Structural Path Models (pSEM, “piecewiseSEM” package, Lefcheck 2016 [↗](#)). pSEM allows modelling direct and indirect causal relationships among a set of response variables and predictors (Shipley 2009 [↗](#)). Further, pSEM uses local estimates of each linear structural equation separately (*i.e.*, parameters are estimated from a series of independent models forming a general causal graph), which allows the inclusion of a large number of parameters despite modest sample sizes (Shipley 2009 [↗](#)). We ran a pSEM for each ecosystem function separately (*i.e.*, seven pSEMs, see an example in **Figure 2** [↗](#)). In each pSEM, the ecosystem function was the dependent variable whereas the six biodiversity estimates (species and genetic diversity estimated for each trophic level) and the two synthetic environmental variables were the predictors. In each model, environmental predictors were allowed to explain each biodiversity estimate (indirect effects of environmental variables through their influence on biodiversity, see **Figure 2** [↗](#)). For some functions (in particular those associated with plant biomass), irrelevant biodiversity-functions links were not included (e.g., the impact of fish or invertebrate diversity on tree biomasses), which results in 34 BEFs (out of the 42 possible links) having been included in the meta-regression (see hereafter).

From each pSEM model, we retrieved the local parameter (standardized estimate, an equivalent to a coefficient of correlation) associated with the direct effect of each biodiversity estimate (six per function, but for some functions for which ecology-irrelevant BEFs were excluded) on the function (coloured arrows in **Figure 2** [↗](#)), which provides both the magnitude and the direction of each BEF. To smoothen comparison, we calculated a standardized effect size for each BEF by applying the Fisher’s Z transformation (Z_r) to the standardized estimates. Positive Z_r indicate positive associations between biodiversity and ecosystem functions, whereas negative Z_r indicate negative relationships. The higher the absolute value of Z_r , the higher the strength of the association. Z_r

Table 1.

Characteristics of the two first principal components identified by the Principal Component Analysis (PCA) ran on the thirteen environmental variables.

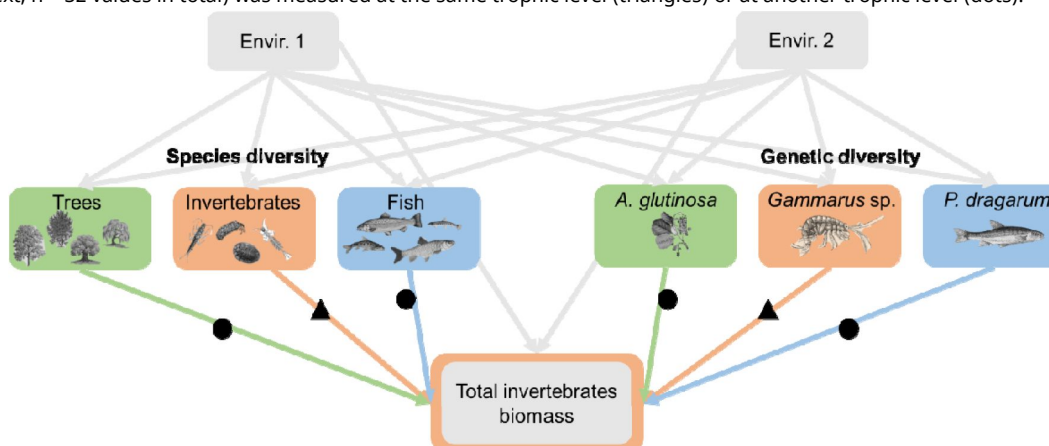
The part of the total environmental variance (%) and the contribution of each variable on each component are shown. The variables that contributed significantly to the axis are highlighted in bold.

	Component 1	Component 2
Part of total Variance (%)	21.66	16.37
River width	0.596	0.320
Connectivity	-0.155	0.646
Altitude	0.648	-0.385
Distance from outlet	0.528	-0.331
East-west gradient	0.105	-0.795
Oxygen concentration	0.738	0.343
Oxygen saturation	0.266	-0.509
Water temperature	-0.594	-0.012
Specific conductivity	-0.463	-0.045
pH	0.573	0.398
Concentration in $\text{NO}_3^+/\text{NO}_2^-$	-0.369	-0.286
Concentration in NH_4^+	-0.019	-0.207
Concentration in PO_4^{3-}	-0.279	0.236
Global characteristic	Low altitude, poorly oxygenated site - High altitude, highly oxygenated sites	Poorly connected east site - Highly connected west site

Figure 2.

Example of one of the seven causal models used to quantify the relationships between (species and genetic) diversity and ecosystem functions.

We focused on seven ecosystem functions associated with genetic and species diversity at three trophic levels (green for primary producer, orange for primary consumer and blue for secondary consumer). Each relationship between biodiversity and ecosystem functions ($n = 6$ values per function, but for some functions for which irrelevant links were not considered, see the text, $n = 32$ values in total) was measured at the same trophic level (triangles) or at another trophic level (dots).



therefore indicate both the direction (positive or negative) and the magnitude of the associations. Our seven measures of ecosystem functions were not correlated one to each other (all $r_{\text{pearson}} < |0.39|$).

Direction and magnitude of all types of BEFs

We used a linear mixed-model to test (i) whether the magnitude and direction of *genetic* BEFs are similar to those of *species* BEFs, and (ii) whether *within-trophic level* BEFs are similar in effect size to *between-trophic level* BEFs. In this model, Z_r (providing the direction and magnitude of each BEF, $n=34$) was the dependent variable, and the predictors were the diversity facets used to measure biodiversity (genetic or species diversity) and the type of BEF (*within-trophic* or *between-trophic* levels, triangles vs. dots in **Figure 2**). We included the two-term interaction between diversity facet and type of BEF to test whether the magnitude and direction of *genetic* and *species* BEFs are consistent across *within-trophic level* and *between-trophic level* BEFs. We further included in this model the type of ecosystem function as a random term (to take into account that each ecosystem function was associated with several biodiversity estimates) as well as the inverse of the asymptotic variance ($v_z=n-3$) associated with each effect size as a weighting parameter for each case study (Balvanera *et al.* 2006; Raffard *et al.* 2019).

We ran an additional linear mixed-model similar to the previous one, except that we added as a fixed effect the trophic level at which biodiversity was measured to estimate BEFs (primary producers, primary consumers or secondary consumers) as well as all interaction terms. Interaction terms allow testing the consistency of major conclusions across trophic levels, thereby determining the extent to which our findings can be generalized along the trophic chain. Models were run using the `lmer` function (“lme4” package) and significance of fixed effects was determined using type III ANOVA with Wald chi-square tests (function `Anova` from the “car” R package, $\alpha=0.05$).

Results

Details of causal models linking environmental parameters, species and genetic diversity and ecosystem functions are graphically depicted in **Figure 3a-g**. The percentage of variance in ecosystem function explained by the environment and biodiversity varies from 10% (invertebrate biomass, **Figure 3e**) to 55% (*Phoxinus* biomass, **Figure 3f**) and was moderate overall. For all functions but the three biomass, part of the variance was (directly) explained by at least one out of the two environmental PCA axes. For some functions (e.g., *Phoxinus* biomass, **Figure 3f**), there was a combined effect of several biodiversity estimates, whereas for other functions (e.g., *Alnus* biomass, Litter decomposition, **Figure 3a**, **3c**) the effect of a single biodiversity estimate predominates. Overall, direct environmental effects on ecosystem functions did not predominate, and environmental effect sizes were similar (in strength) to that of biodiversity effects, showing the important role of biodiversity for ecosystem functions in the wild.

Individual effect sizes (Z_r) measured between biodiversity estimates and ecosystem functions were weak to moderate, irrespectively of the considered ecosystem function and of the type of BEFs (*genetic/species* BEFs, *within-trophic level/ between-trophic level* BEFs) (**Figure 4a**, **4b**, Table S2). As expected under natural conditions (Hagan *et al.* 2021), BEFs ranged from negative to positive, and their distribution were centred around 0, although we observed a slight tendency for *genetic* BEFs toward positive values (**Figure 3b**). Four out of the 34 BEFs were strong and significant; two significant BEFs concerned *species* BEFs (negative relationship between the biomass of *A. glutinosa* and the diversity of trees, $Z_r = -0.446$, 95% CI [-0.695, -0.143]; negative relationship between the biomass of *P. dragarum* and the diversity of fish, $Z_r = -0.529$, 95% CI [-0.802, -0.166]) and two concerned *genetic* BEFs (negative relationship between the biomass of *P. dragarum* and the diversity of *A. glutinosa*, $Z_r = -0.321$, 95% CI [-0.602, -0.019]; positive relationship

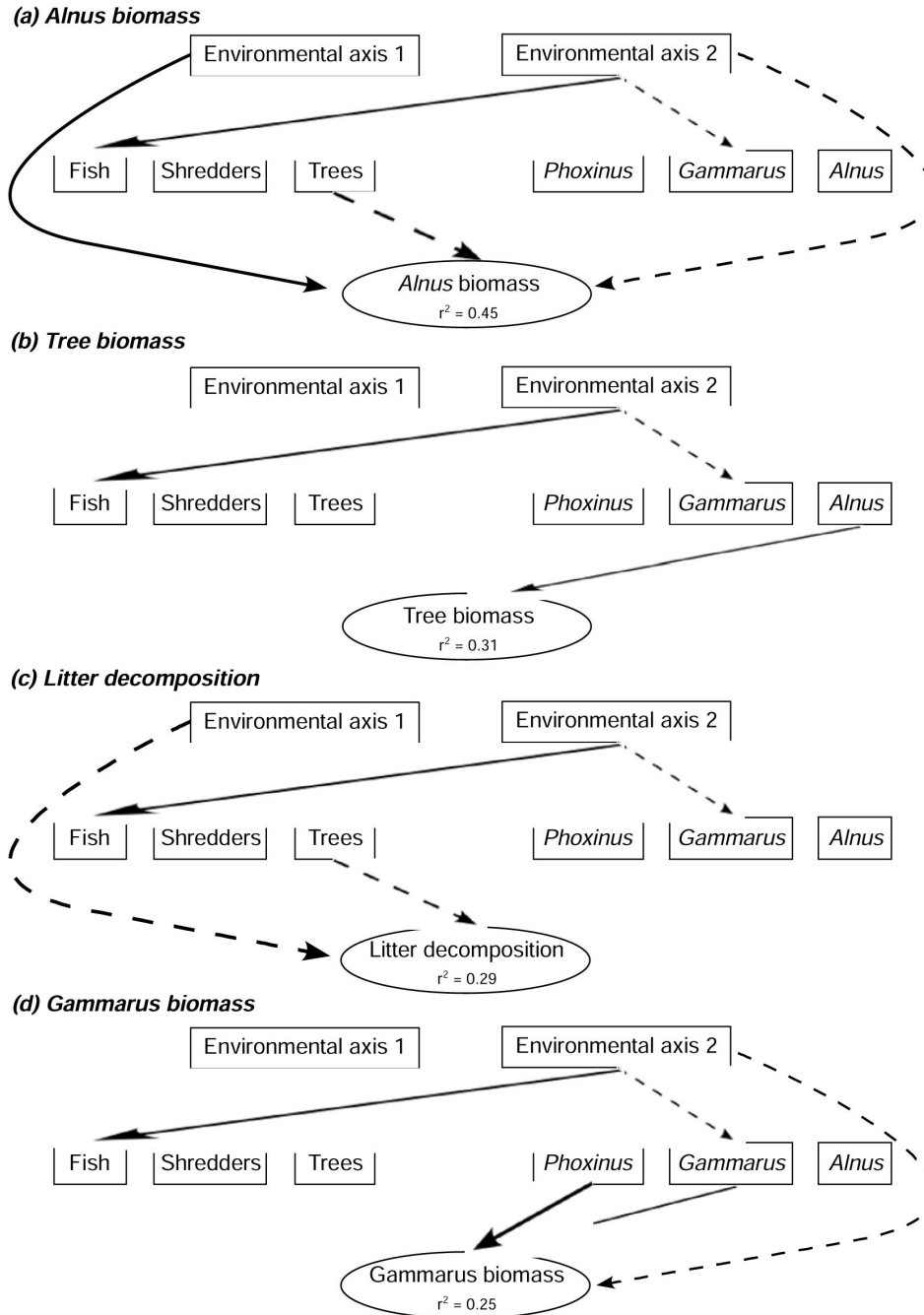


Figure 3.

Details of the seven causal models linking abiotic parameters, species and genetic diversity and ecosystem functions.

Each causal graph (a to g) represents a simplified illustration of the relationships between the two PCA axes synthesizing the environmental parameters of each sampling site (Environmental axis 1 and 2), the species diversity estimated at each trophic level (boxes “Fish”, “Shredders” and “Trees”), the genomic diversity estimated from each focal species at each trophic level (boxes “*Phoxinus*”, “*Gammarus*” and “*Alnus*”), and each ecosystem function (one model per function). Only the relationships for which the p-value was inferior to 0.20 are indicated for visual simplification. Full arrows indicated positive effects, whereas dotted arrows indicated negative effects. The width of the arrows is proportional to the size of their effects. The percentage of variance explained by environmental and biodiversity effects on ecosystem functions (r^2) is indicated for each function.

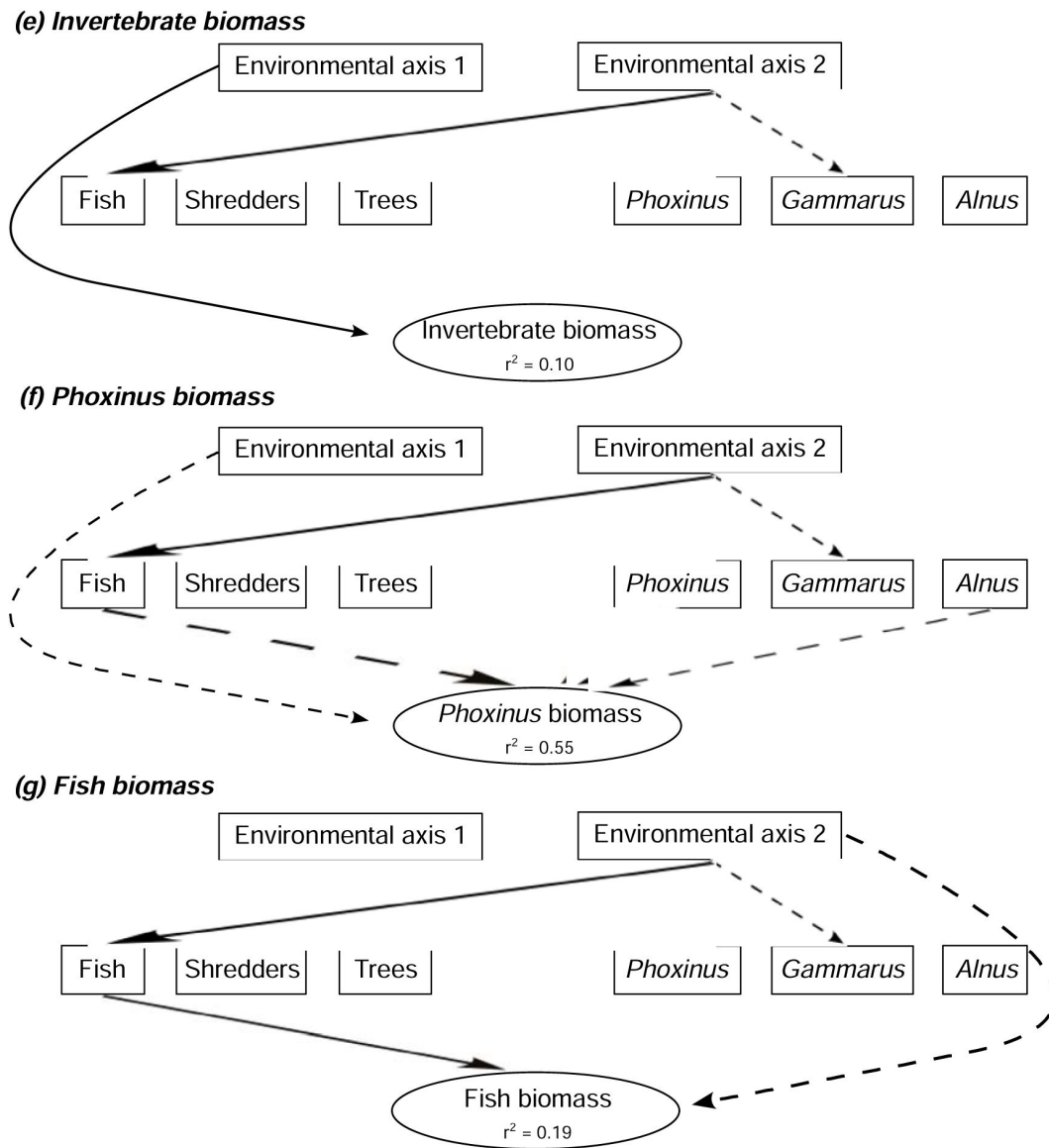


Figure 3. (continued)

between the biomass of *Gammarus* sp. and the diversity of *P. dragarum*, $Zr = 0.446$, 95% CI [0.001, 0.829]) (**Figure 4a** [↗](#)). Noteworthy, for *within-trophic* BEFs, most case studies fall into the category whereby genetic BEFs tend to be positive and species BEFs tend to be negative (grey bottom-right square in **Figure 4a** [↗](#)).

We confirmed this visual tendency by summarizing all individual Zr through a meta-regression. Indeed, we found a significant interaction between the facet at which biodiversity is measured (genetic or species diversity), and the type of BEF that was measured (within- or between trophic levels; **Table 2** [↗](#)). This interaction indicates (i) that -overall-*within-trophic level* BEFs were significantly negative when considering species diversity ($Zr_{\text{Within*Species}} = -0.185$, 95% CI [-0.343, -0.027]), whereas *within-trophic level* BEFs were significantly positive when considering genetic diversity ($Zr_{\text{Within*Genetic}} = 0.168$, 95% CI [0.010, 0.326], see **Figure 5a** [↗](#)), and (ii) that this pattern was not observed for *between-trophic levels* BEFs, where no particular trend was observed (**Figure 5a** [↗](#)). Although most individual Zr were weak to moderate (and not significant), their consistency (in term of magnitude and direction) resulted in a significant pattern whereby species and genetic diversity have opposite effects on ecosystem functions for *within-trophic level* BEFs; species diversity is negatively associated, whereas genetic diversity is positively associated with ecosystem functions, but only when the influence of biodiversity on ecosystem functions is measured within the same trophic level.

When including the trophic level at which biodiversity is measured, we found no significant interaction terms between trophic levels and other fixed effects nor any additive effect of trophic levels (see **Table S1** [↗](#)). This indicates that our main findings were consistent across trophic levels, *i.e.*, the respective negative and positive effects on ecosystem functions of species and genetic diversity hold statistically true across all trophic levels (**Figure 5b** [↗](#)).

Discussion

We provide empirical evidence that, in natural ecosystems, the effect sizes of genetic and species diversity on multi-trophic ecosystem functions are of similar magnitude, but operate in opposite directions. Indeed, for BEFs measured within the same trophic level, the effects of species diversity across multiple ecosystem functions were moderately negative on average, whereas the effects of genetic diversity were moderately positive. This suggests an antagonistic effect between the genetic and the species components of biodiversity in the modulation of ecosystem functions within one trophic level. This antagonistic effect was not identified for BEFs measured across trophic levels, since in these cases the influence of both genetic diversity and species diversity across multiple ecosystem functions was generally not different from zero. These conclusions hold true across three trophic levels (plants, invertebrates and fish), indicating that the relative effects of genetic and species diversity on ecosystem functions are not limited to a specific trophic level.

Our study is one of the few field-based study revealing BEFs across an entire (riverine) food chain spanning from primary producers to secondary consumers. Indeed, most previous BEF studies in the field focused on a single trophic level, and predominantly on terrestrial primary producers (Duffy *et al.* 2017 [↗](#); Van Der Plas 2019 [↗](#), but see *e.g.*, Li *et al.* 2020 [↗](#), Moi *et al.* 2021 [↗](#)). This permitted encompassing a broad range of ecosystem functions that depict the overall functioning of a riverine ecosystem (rather than focusing on a single compartment). Moreover, we focused both on the effects of genetic and species diversity on these ecosystem functions, which has rarely (if not ever) been evaluated so far and which provides an exhaustive understanding of BEFs in the wild. Our causal analyses also took into account the direct (and indirect) effect of environmental factors on ecosystem functions, which is a prerequisite to isolate biodiversity effects. Direct environmental effects on ecosystem functions were (in average) not stronger in intensity than biodiversity effects, which is coherent with previous syntheses on species BEFs in the wild (Duffy *et al.* 2017 [↗](#)). Furthermore, environmental factors used to describe sampling sites in this study

Figure 4.

General description of individual effect sizes measured between biodiversity estimates and ecosystem functions (BEFs) in a riverine trophic chain.

(a) The magnitude and direction of individual effect sizes (Z_r) of biodiversity is shown for each ecosystem functions as a biplot between Z_r associated with genetic diversity (y-axis, *genetic* BEFs) measured for one of three target species (*Alnus glutinosa*, *Gammarus* sp. and *Phoxinus phoxinus*) and Z_r associated with species diversity (x-axis, *species* BEFs) measured for one of three trophic levels (trees, invertebrates and fish). For each ecosystem function (but the biomass of trees and of *A. glutinosa*), a total of six Z_r are depicted in the biplot; four of them are associated with biodiversity measured at another trophic level than the one of the target functions (red symbols, e.g., effect of fish diversity on invertebrate biomass) and two of them are associated with biodiversity measured at the same trophic level than the one of the target functions (blue symbols, e.g., effect of fish diversity on fish biomass). The arrows indicate significant Z_r (95% confidence intervals excluded 0, see Table S2); vertical arrows are for significant *genetic* BEFs, horizontal arrows are for *species* BEFs. White quadrants stand for situation in which genetic and species BEFs are in the same direction, whereas grey quadrants indicate situation in which genetic and species BEFs are in the opposite direction. Within each quadrant, sub-quadrants indicate the relative magnitude of BEFs, i.e., whether *genetic* BEFs are stronger, weaker or equal in magnitude than *species* BEFs. (b) Density plots displaying the distribution of individual Z_r for *species*- and *genetic* BEFs (dotted and full lines respectively).

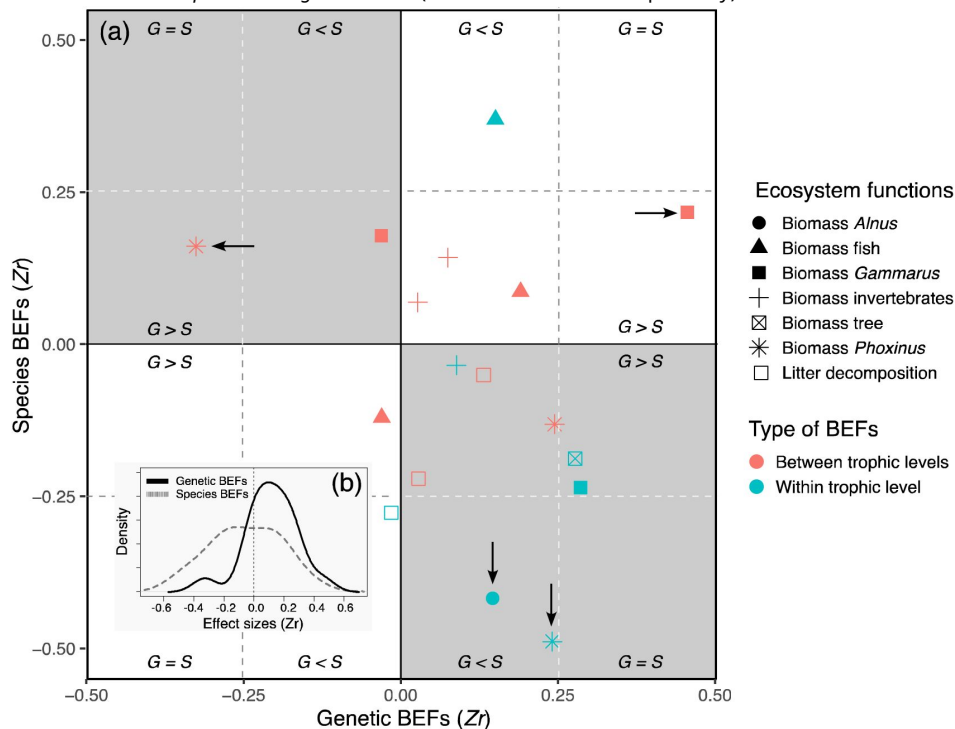


Table 2.

ANOVA table for the linear mixed model testing whether the relationships between biodiversity and ecosystem functions measured in a riverine trophic chain differ between the biodiversity facets (species or genetic diversity) and the types of BEF (*within*- or *between*-trophic levels). A Wald chi-square test is used to test the significance of each fixed effect.

	Degree of freedom	Chisq-value	P-value
(Intercept)	1	0.287	0.595
Biodiversity facet	1	0.232	0.630
Type of BEF	1	5.393	0.020
Biodiversity facet*Type of BEF	1	5.567	0.018

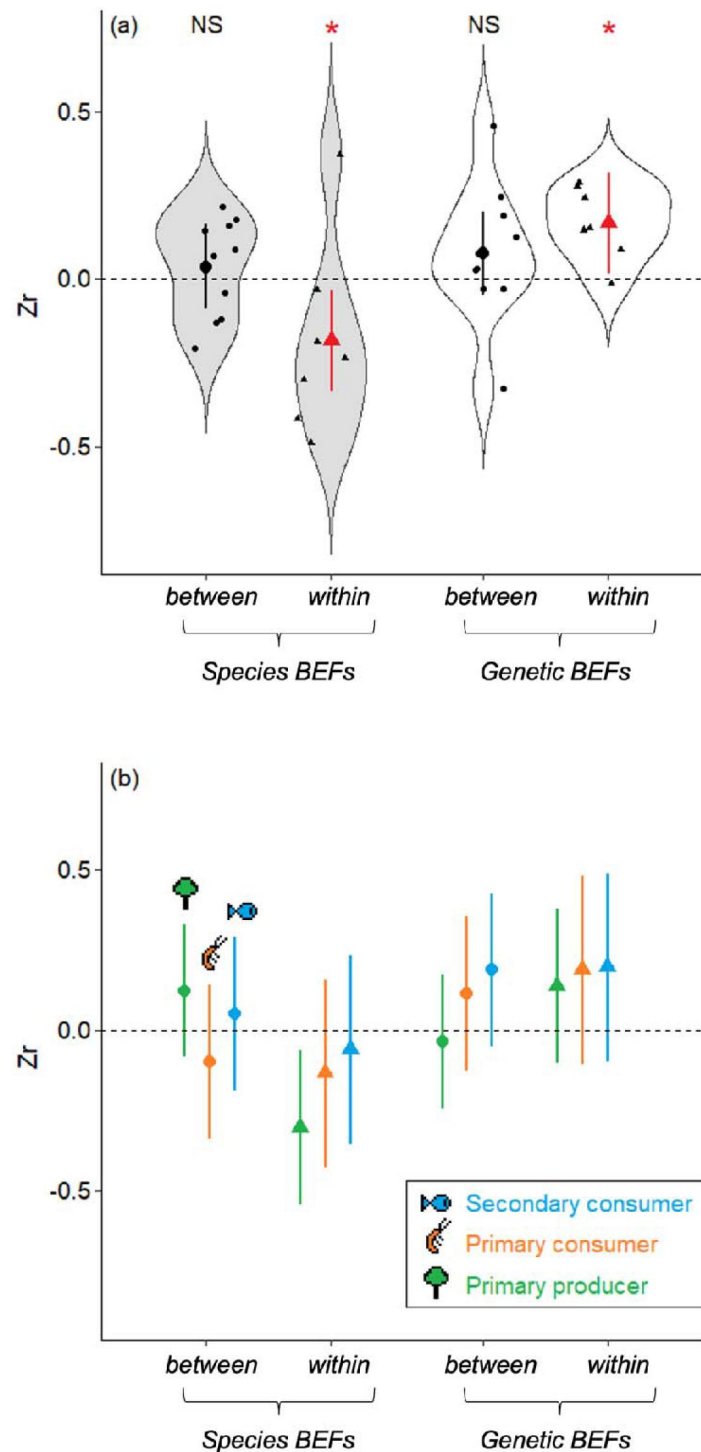


Figure 5.

Magnitude and direction of the mean effects sizes estimated from the relationships between biodiversity and ecosystem functions (BEFs) measured in a riverine trophic chain.

(a) The magnitude and direction of BEFs are expressed as effect sizes (Zr) and are displayed according to the facet used to measured biodiversity (genetic or species diversity, light grey and white boxplots respectively) and to the type of BEFs (*within-trophic level* BEFs or *between-trophic level* BEFs, triangles and dots respectively). Red colour and stars indicate global effect sizes that are significantly different from zero (p-value < 0.05). Large symbols are mean $\pm$ 95%IC. (b) Same representation as (a) but with details at each trophic level (mean $\pm$ 95%IC, green for primary producer, orange for primary consumer and blue for secondary consumer). The trophic level at which BEFs are measured is coherent across all trophic levels.

were not strong predictors of species and genetic biodiversity. Two non-exclusive hypotheses may explain this observation: (i) using PCA axes to resume environmental gradients may blur some specific environment-biodiversity links, and (ii) as shown and explained in a companion paper (Fargeot *et al.* 2023 [↗](#)), the East-West gradient used in this study (rather than a classical upstream-downstream gradient) intrinsically limits the potential for strong environmental effects on biodiversity (which was the purpose of this sampling design). Nonetheless, after accounting for these environmental covariates, we found that most individual BEFs (either *genetic* or *species*, *within-trophic levels* or *between-trophic levels* BEFs) were weak to moderate in magnitude, and that they operated almost equally in both direction (i.e., positive and negative association between biodiversity and ecosystem functions). As such, the distribution of individual effect sizes was centred around 0, for both *genetic* and *species* BEFs. Accordingly, there were only four individual BEFs that were significant, three out of them were negative and one was positive (Table S2 [↗](#)). This general pattern (low to moderate BEFs with both positive and negative direction) is actually consistent with the most exhaustive meta-analysis having synthesized the magnitude and direction of *species* BEFs in the wild (Van Der Plas 2019 [↗](#)) and with recent conceptual works (Hagan *et al.* 2021 [↗](#)) concluding that strong and positive BEFs should not be the norm in natural ecosystems, but rather that a mix of positive, neutral and negative BEFs are expected. Our results confirm, using a dataset encompassing three trophic levels in a single ecosystem type, this conclusion.

We focused both on *within-trophic level* and *between-trophic level* BEFs, which likely encompasses a broad array of mechanisms sustaining potential associations between biodiversity and ecosystem functions. For instance, two out of the significant BEFs we reveal are negative association between species diversity (fish or tree species diversity respectively) and the biomass production of one of the target species (*Phoxinus sp.* and *Alnus sp.* respectively). These *within-trophic level* BEFs can -for instance-arise either because, if resources are limited, increased number of species within a patch limit the biomass production of each individual species, or because of a poorer competitive ability of the target species under some environmental conditions, which favors the settlement of additional species. Although teasing apart these two hypotheses is difficult, the former hypothesis implies a negative co-variation within the biomass of the focal species and that of the total pool of species, which was not observed, neither for fish, not for trees (results not shown). The other two significant BEFs concerned the association (either positive or negative) between the genetic diversity of a target species (*Alnus sp.* or *Phoxinus sp.*) with the biomass production of another target species (*Phoxinus sp.* or *Gammarus sp.*, respectively). These *between-trophic level* BEFs likely arise through indirect effects implying the diversity and availability of (prey) resources. An obvious limit of this field-based study is the impossibility to tease out these mechanisms. Another limit is associated with the fact that, although environmental covariates were taken into account in causal models, they were synthesized by two PCA axes, and we cannot ensure that all potential environmental covariates have been taken into account. This can influence the actual estimates of BEFs in the wild (Duffy *et al.* 2017 [↗](#)). Nonetheless, it is noteworthy that we have previously shown that -in this dataset- species and genetic diversity were not correlated one to each other and that each biodiversity facet was sustained by different environmental predictors (Fargeot *et al.* 2023 [↗](#), see also Figure 3 [↗](#)). This implies that environmental and biodiversity effects inferred in this study are relatively robust (i.e., not distorted by strong collinearities) and can be interpreted independently one from each other (e.g., the positive effect *Phoxinus* genetic diversity on *Phoxinus* biomass is independent from the negative of fish species diversity as the two estimates of biodiversity do not co-vary, see Figure 3f [↗](#)).

Keeping limitations associated with field-based studies into account (see above), revealing associations between ecosystem functions and species and genetic biodiversity (or the lack of) in natural ecosystems is -as ecologists-an important step forward to set theoretical and experimental approaches aiming at understanding this complex biological reality. Beyond individual BEF case studies (that were not the main aim of this study), their aggregation across trophic levels and

biodiversity facets revealed a clear (and statistically supported) pattern whereby, within trophic levels, genetic and species diversity display antagonistic association with ecosystem functions; the global effect of species diversity across multiple ecosystem functions was negative, whereas the global effect of genetic diversity was positive. This pattern emerges from the “cumulative” effects of weak to moderate associations between biodiversity and ecosystem functions that consistently point toward the same direction (positive for genetic diversity, negative for species diversity), emphasizing the meaningfulness of meta-regressions (and more generally approaches based on effect sizes rather than on p-values) to reveal biological patterns. We hereafter discuss the ecological relevance of this general pattern.

Our results confirm a previous meta-analysis demonstrating that genetic and species diversity modulates ecosystem functions with a similar magnitude (Raffard *et al.* 2019 [DOI](#)), and results from few experimental studies that manipulated both the genetic and species components of biodiversity under controlled conditions (e.g., Jiang *et al.* 2022 [DOI](#); Prieto *et al.* 2015 [DOI](#)). Indeed, within trophic levels, the absolute mean effect size of genetic and species diversity across ecosystem functions were of the same magnitude ($|Zr| = 0.168$ and 0.185 for genetic and species diversity effects respectively), and slightly greater than the effect sizes reported under controlled conditions ($|lnRR| = 0.132$ and 0.134 for genetic and species diversity effects respectively, Raffard *et al.* 2019 [DOI](#)) and those more generally reported for *species* BEFs ($|Zr| = 0.101$, Balvanera *et al.* 2006 [DOI](#)). This demonstrates for the first time that under natural conditions, the effects of genetic and species components of biodiversity on ecosystem functions are comparable. However, our study goes two steps further as (i) it extends the conclusion made by Raffard *et al.* (2019) [DOI](#) to multiple trophic levels and (ii) it demonstrates that the effects of genetic and species BEFs can actually operate in opposite directions.

As pointed out by Raffard *et al.* (2019) [DOI](#), the vast majority (91% of 23 reviewed studies by 2019, see also Wan *et al.* 2022 [DOI](#)) of studies investigating the effects of genetic diversity on ecosystem functions have focused on primary producers, and all of them were based on experiments, which is also the case for most studies manipulating both genetic and species diversity. These trends strongly hamper any generalization. On the contrary, our findings provide a strong support for broadening the conclusion that both genetic and species diversity can influence ecosystem functions in the wild. More strikingly, our results demonstrate that, although the absolute effect sizes of genetic and species BEFs are of similar magnitude, for *within-trophic level* BEFs, the direction of their effects are opposite; species diversity (in general) reduces the rate of ecosystem functions, whereas genetic diversity enhances the same functions. For instance, all other things being equal, higher fish species diversity is associated with a lower productivity (biomass) in *P. dragarum* (see above for potential explanations), whereas its own genetic diversity tends to be associated with a higher productivity (Table S2 [DOI](#)). As genetic and species diversity have been found to be uncorrelated spatially in this landscape (Fargeot *et al.* 2023 [DOI](#)), covariation among diversity estimates cannot explain these patterns. These antagonistic effects of genetic and species diversity on ecosystem functions parallel previous experimental findings on plants (Hazard *et al.* 2017 [DOI](#); Tang *et al.* 2022 [DOI](#)). It is now essential to understand the mechanisms sustaining these antagonistic effects as a step forward.

Species BEFs were on average negative (see Table S2 [DOI](#) for individual estimates), which contrasts with the general view that species biodiversity favours ecosystem functions, although it is not that surprising (Dee *et al.* 2023 [DOI](#); Hagan *et al.* 2021 [DOI](#)). Indeed, the net effect of species biodiversity on ecosystem functions results from the combined effects of both negative factors, arising from antagonistic interactions such as negative complementarity or negative selection effect, and positive factors, arising from beneficial interactions such as niche complementarity or facilitation (Loreau & Hector 2001 [DOI](#)). It is likely that, in our case, the net effect of interspecific interactions mostly results from negative complementarity among species (or strong negative selection effect), whereas the net effect of intraspecific interactions may result from facilitative interactions and/or improved niche complementarity with increased genetic diversity. Intraspecific competition is

generally stronger than interspecific competition (Connell 1983 [↗](#)), and intraspecific interactions could be expected to lead more frequently to negative complementarity (and hence negative *genetic* BEFs) than interspecific interactions. Since we observe the opposite, we can hypothesize that genetic diversity is essential to increase niche complementarity within species (Bolnick *et al.* 2003 [↗](#)) and hence to reduce the pervasive effects of intraspecific interactions (Hughes *et al.* 2008 [↗](#); Prunier *et al.* 2023 [↗](#)). Given the empirical nature of our study, it is hard to tease apart underlying mechanisms. Theoretical approaches, modelling simultaneously the genetic and species components of biodiversity, would be extremely useful to reveal the mechanisms sustaining opposite effects of intra- and interspecific diversity on ecosystem functions.

These antagonistic effects were observed only for BEFs measured *within* trophic levels, not for those measured *between* trophic levels. An overall *between-trophic level* BEF not different from zero suggests that biodiversity at a trophic level has only limited impact on ecosystem functions at another trophic level. For example, the biomass of *P. dragarum* was primarily influenced by genetic and species diversity in fishes, rather than the diversity of their preys (Table S2 [↗](#)). However, for both genetic and species estimates of biodiversity, there was a substantial variation in effect sizes for *between-trophic level* BEFs that ranged from negative to positive BEFs (Figure 5 [↗](#)). This suggests that biodiversity effects across trophic levels may be more variable in their direction than within-trophic level BEFs, which appear as more constrained. Variability in the magnitude and direction of effect sizes for *between-trophic level* BEFs likely blur a more general trend, but this variation is actually expected under natural conditions in which interactions involve multiple prey and predator species, fostering co-adaptation among communities from different trophic levels (Aubree *et al.* 2020 [↗](#); Poisot *et al.* 2013 [↗](#)). In these cases, trophic complementarity between two trophic levels (i.e., the originality of a species based on the identity of the species it interacts with) might be a stronger determinant of ecosystem functions than complementarity measured at either one of the two trophic levels (Poisot *et al.* 2013 [↗](#)). Quantifying trophic complementarity among our three target species (and communities) using stable isotope or gut content analyses for instance would be extremely valuable to assess whether this complexity can better explain BEFs between trophic levels than diversity measured at one of the trophic level (Aubree *et al.* 2020 [↗](#)).

The empirical patterns we revealed here were all extremely consistent across the three trophic levels, hence allowing generalization. It is noteworthy that, although statistically strong and consistent, these patterns must be interpreted with care as field-based approaches are limited in properly taking into account the environmental heterogeneity of natural ecosystems (Hagan *et al.* 2021 [↗](#)). BEFs were not particularly stronger at any specific trophic level and the relative effects of genetic and species diversity were not dependent on the trophic level at which the function was estimated. We may have expected a stronger top-down regulation (i.e., biodiversity of predators has more effects than biodiversity of preys) of ecosystem functions since previous studies showed that biodiversity loss should have greater consequences for multi-functionality when it occurs at higher trophic levels (Lefcheck *et al.* 2015 [↗](#), Seibold *et al.* 2018 [↗](#)). For instance, increased genetic diversity within a predatory fish species has experimentally been shown to indirectly increase the rate of litter decomposition by increasing the diversity of shredders (Raffard *et al.* 2021 [↗](#)). Similarly, the relative effects of genetic and species diversity on functions may have varied among trophic levels, and in particular the relative importance of genetic diversity may have been higher for species-poor trophic levels (i.e., fish community) because of a “compensatory effect”. We found no evidence for these potential trophic-level dependencies, but instead found extremely consistent patterns, which, from a broader perspective, reveal the importance of integrating both multi-trophic and multi-faceted approaches in predicting the overall consequences of biodiversity loss on ecosystem functioning.

To conclude, we found that the genetic (intraspecific) and species (interspecific) facets of biodiversity are both important drivers of multiple ecosystem functions in a natural and multi-trophic context. In the wild, these two facets of biodiversity can, as expected, generate low to

moderately high impacts on ecosystem functions measured across three trophic levels, and they can operate in opposite directions. This demonstrates the importance for managers to develop integrative conservation plans spanning the entire diversity of life (from genes to species). For instance, genetic diversity loss often precedes species loss, and our results suggest that -in mountain streams-losing genes may actually be particularly detrimental for the performance of ecosystem functions. As such, it appears essential to maintain populations with high levels of genetic diversity in these ecosystems. Future studies should (i) extend these findings to other ecosystems and by quantifying natural genetic variation in more than a single species per trophic level, (ii) generate theoretical predictions regarding the mechanisms sustaining the antagonistic effects of genetic and species diversity on functions we revealed, and (iii) use a broader integrative approach for estimating biodiversity across facets (*inclusive* biodiversity) by using either a trait-based approach or a genetic-based approach as recently proposed by Blanchet *et al.* (2023) [and](#) Loreau *et al.* (2023) [and](#).

Author contribution

Laura Fargeot: Conceptualization (Supporting); Methodology (Equal); Software (Equal); Validation (Equal); Formal analysis (Lead); Investigation (Lead); Data curation (Lead); Writing - original draft (Lead); Writing - review & editing (Equal); Visualization (Lead); Supervision (Equal); Project administration (Lead). Camille Poesy: Methodology (Equal); Validation (Equal); Investigation (Lead); Data curation (Supporting); Supervision (Equal); Project administration (Supporting). Maxim Lefort: Methodology (Supporting); Investigation (Lead); Data curation (Supporting); Supervision (Equal); Project administration (Supporting). Jérôme G. Prunier: Methodology (Supporting); Software (Equal); Formal analysis (Supporting); Investigation (Supporting); Resources (Equal); Data curation (Supporting); Writing - original draft (Supporting); Writing - review & editing (Supporting). Madoka Krick: Methodology (Supporting); Investigation (Equal). Rik Verdonck: Methodology (Supporting); Software (Equal); Investigation (Supporting); Writing - review & editing (Supporting). Charlotte Veyssière: Methodology (Supporting); Investigation (Equal). Murielle Richard: Methodology (Supporting); Investigation (Supporting); Writing - review & editing (Supporting). Delphine Legrand: Validation (Equal); Writing - review & editing (Equal); Visualization (Supporting). Géraldine Loot: Validation (Equal); Investigation (Supporting); Writing - review & editing (Supporting); Visualization (Supporting). Simon Blanchet: Conceptualization (Lead); Methodology (Lead); Software (Supporting); Validation (Equal); Formal analysis (Supporting); Investigation (Lead); Resources (Equal); Data curation (Supporting); Writing - original draft (Supporting); Writing - review & editing (Lead); Visualization (Lead); Supervision (Lead); Project administration (Lead); Funding acquisition (Lead).

Supplementary materials

	Degree of freedom	Chisq-value	P-value
(Intercept)	1	1.453	0.228
Biodiversity facet	1	1.293	0.255
Type of BEF	1	7.498	0.006
Trophic level	2	1.976	0.372
Biodiversity facet*Type of BEF	1	7.884	0.005
Biodiversity facet*Trophic level	2	3.520	0.172
Trophic level*Type of BEF	2	2.901	0.234
Biodiversity facet*Type of BEF*Trophic level	1	2.994	0.224

Table S1.

ANOVA table for the linear mixed model testing whether the relationships between biodiversity and ecosystem functions measured in a riverine trophic chain differ between the biodiversity facets (species or genetic diversity), the types of BEF (*within- or between-trophic levels*) and the trophic levels at which BEFs are estimated (primary producers, primary consumers or secondary consumers). A Wald chi-square test is used to test the significance of each fixed effect.

Ecosystem function	Predictor	Biodiversity facet	Type of BEF	Zr	negative 95% CI	positive 95% CI	P values
<i>Phoxinus</i> biomass	Fish species diversity	Species diversity	Within-trophic levels	-0.529	-0.803	-0.166	0.006
<i>Alnus</i> biomass	Tree species diversity	Species diversity	Within-trophic levels	-0.447	-0.695	-0.143	0.006
<i>Phoxinus</i> biomass	<i>Alnus</i> genetic diversity	Genetic diversity	Between-trophic levels	-0.321	-0.602	-0.020	0.045
Litter decomposition	Tree species diversity	Species diversity	Within-trophic levels	-0.293	-0.630	0.060	0.115
<i>Gammarus</i> biomass	Invertebrate species diversity	Species diversity	Within-trophic levels	-0.273	-0.661	0.128	0.194
Litter decomposition	Fish species diversity	Species diversity	Between-trophic levels	-0.210	-0.625	0.210	0.337
Tree biomass	Tree species diversity	Species diversity	Within-trophic levels	-0.183	-0.485	0.122	0.250
Fish biomass	Invertebrate species diversity	Species diversity	Between-trophic levels	-0.138	-0.428	0.153	0.506
<i>Phoxinus</i> biomass	Invertebrate species diversity	Species diversity	Between-trophic levels	-0.129	-0.471	0.215	0.362
Invertebrate biomass	Invertebrate species diversity	Species diversity	Within-trophic levels	-0.051	-0.381	0.279	0.766
Fish biomass	<i>Gammarus</i> genetic diversity	Genetic diversity	Between-trophic levels	-0.047	-0.457	0.364	0.825
<i>Gammarus</i> biomass	<i>Alnus</i> genetic diversity	Genetic diversity	Between-trophic levels	-0.043	-0.438	0.352	0.833
Litter decomposition	Invertebrate species diversity	Species diversity	Between-trophic levels	-0.020	-0.401	0.361	0.918
Litter decomposition	<i>Alnus</i> genetic diversity	Genetic diversity	Within-trophic levels	-0.012	-0.394	0.370	0.950
Litter decomposition	<i>Phoxinus</i> genetic diversity	Genetic diversity	Between-trophic levels	0.018	-0.379	0.416	0.930
Invertebrate biomass	<i>Alnus</i> genetic diversity	Genetic diversity	Between-trophic levels	0.025	-0.306	0.356	0.885
Invertebrate biomass	Tree species diversity	Species diversity	Between-trophic levels	0.063	-0.236	0.362	0.682
Invertebrate biomass	<i>Phoxinus</i> genetic diversity	Genetic diversity	Between-trophic levels	0.085	-0.260	0.429	0.634
Fish biomass	Tree species diversity	Species diversity	Between-trophic levels	0.087	-0.276	0.449	0.644
Invertebrate biomass	<i>Gammarus</i> genetic diversity	Genetic diversity	Within-trophic levels	0.097	-0.242	0.435	0.580
Litter decomposition	<i>Gammarus</i> genetic diversity	Genetic diversity	Between-trophic levels	0.110	-0.281	0.500	0.586
<i>Alnus</i> biomass	<i>Alnus</i> genetic diversity	Genetic diversity	Within-trophic levels	0.128	-0.179	0.433	0.422
Invertebrate biomass	Fish species diversity	Species diversity	Between-trophic levels	0.144	-0.219	0.504	0.444
Fish biomass	<i>Phoxinus</i> genetic diversity	Genetic diversity	Within-trophic levels	0.159	-0.260	0.575	0.464
<i>Gammarus</i> biomass	Tree species diversity	Species diversity	Between-trophic levels	0.161	-0.198	0.516	0.388
<i>Phoxinus</i> biomass	Tree species diversity	Species diversity	Between-trophic levels	0.165	-0.100	0.426	0.233
<i>Gammarus</i> biomass	Fish species diversity	Species diversity	Between-trophic levels	0.185	-0.249	0.614	0.413
Fish biomass	<i>Alnus</i> genetic diversity	Genetic diversity	Between-trophic levels	0.193	-0.211	0.592	0.360
<i>Phoxinus</i> biomass	<i>Gammarus</i> genetic diversity	Genetic diversity	Between-trophic levels	0.228	-0.074	0.522	0.151
<i>Phoxinus</i> biomass	<i>Phoxinus</i> genetic diversity	Genetic diversity	Within-trophic levels	0.244	-0.064	0.542	0.133
Tree biomass	<i>Alnus</i> genetic diversity	Genetic diversity	Within-trophic levels	0.280	-0.062	0.609	0.121
<i>Gammarus</i> biomass	<i>Gammarus</i> genetic diversity	Genetic diversity	Within-trophic levels	0.292	-0.120	0.688	0.178
Fish biomass	Fish species diversity	Species diversity	Within-trophic levels	0.326	-0.123	0.753	0.169
<i>Gammarus</i> biomass	<i>Phoxinus</i> genetic diversity	Genetic diversity	Between-trophic levels	0.446	0.007	0.830	0.047

Table S2

Estimates of individual effect sizes of BEFs (Zr, n=34) for each ecosystem function, each biodiversity facet (genetic or species diversity) and each type of BEF (within- or between-trophic levels).

95% confidence intervals are provided together with the estimate of each BEF. BEFs are considered as significant when the 95%CI does not overlap 0. P-values estimated from t-test are also provided.

References

- Altermatt F. (2013) **Diversity in riverine metacommunities: a network perspective** *Aquatic Ecology* **47**:365–377
- Aubree F., David P., Jarne P., Loreau M., Mouquet N., Calcagno V. (2020) **How community adaptation affects biodiversity–ecosystem functioning relationships** *Ecol Lett* **23**:1263–1275
- Balvanera P., Pfisterer A.B., Buchmann N., He J.-S., Nakashizuka T., Raffaelli D., et al. (2006) **Quantifying the evidence for biodiversity effects on ecosystem functioning and services: Biodiversity and ecosystem functioning/services** *Ecol Lett* **9**:1146–1156
- Blanchet S., Fargeot L., Raffard A. (2023) **Phylogenetically conserved candidate genes unify biodiversity–ecosystem function relationships and eco|evolutionary dynamics across biological scales** *Mol Ecol* **32**:4467–4481
- Blanchet S., Prunier J.G., Paz|Vinas I., Saint|Pé K., Rey O., Raffard A., et al. (2020) **A river runs through it: The causes, consequences, and management of intraspecific diversity in river networks** *Evol Appl* **13**:1195–1213
- Bolnick D.I., Svanback R., Fordyce J.A., Yang L.H., Davis J.M., Hulsey C.D., et al. (2003) **The ecology of individuals: Incidence and implications of individual specialization** *Am Nat* **161**:1–28
- Cardinale B.J., Duffy J.E., Gonzalez A., Hooper D.U., Perrings C., Venail P., et al. (2012) **Biodiversity loss and its impact on humanity** *Nature* **486**:59–67
- Carnevali M. (2022) **Etude phylogénétique des populations de Gammarus (Gammarus sp.) du piedmont Pyrénéen** *Master thesis* University of Toulouse, France
- Chao A., Chiu C.-H., Jost L. (2014) **Unifying Species Diversity, Phylogenetic Diversity, Functional Diversity, and Related Similarity and Differentiation Measures Through Hill Numbers** *Annu Rev Ecol Evol Syst* **45**:297–324
- Connell J.H. (1983) **On the prevalence and relative importance of interspecific competition|: evidence from field experiments** *Am Nat* **122**:661–696
- Crutsinger G.M., Collins MD, Fordyce J.A., Gompert Z., Nice C., Sanders N.J. (2006) **Plant Genotypic Diversity Predicts Community Structure and Governs an Ecosystem Process** *Science* **313**:966–968
- De Kort H., Baguette M., Prunier J.G., Tessier M., Monsimet J., Turlure C., et al. (2018) **Genetic costructure in a meta-community under threat of habitat fragmentation** *Mol Ecol* **27**:2193–2203
- Dee L.E., Ferraro P.J., Severen C.N., Kimmel K.A., Borer E.T., Byrnes J.E.K., et al. (2023) **Clarifying the effect of biodiversity on productivity in natural ecosystems with longitudinal data and methods for causal inference** *Nat Commun* **14**

- Dray S., Dufour A.-B. (2007) **The ade4 Package: Implementing the Duality Diagram for Ecologists** *J Stat Soft* **22**
- Duffy J.E., Godwin C.M., Cardinale B.J. (2017) **Biodiversity effects in the wild are common and as strong as key drivers of productivity** *Nature* **549**:261–264
- Duffy J.E., Lefcheck J.S., Stuart-Smith R.D., Navarrete S.A., Edgar G.J. (2016) **Biodiversity enhances reef fish biomass and resistance to climate change** *P Natl Acad Sci* **113**:6230–6235
- Fargeot L., Poesy C., Lefort M., Prunier J.G., Krick M., Veyssi re C., et al. (2023) **Genomic and species diversity patterns across multiple trophic levels in riverscapes** *Oikos* **2023**
- Fridley J.D., Grime J.P. (2010) **Community and ecosystem effects of intraspecific genetic diversity in grassland microcosms of varying species diversity** *Ecology* **91**:2272–2283
- Grace J.B., Anderson T.M., Seabloom E.W., Borer E.T., Adler P.B., Harpole W.S., et al. (2016) **Integrative modelling reveals mechanisms linking productivity and plant species richness** *Nature* **529**:390–393
- Griesmann M., Chang Y., Liu X., Song Y., Haberer G., Crook M.B., et al. (2018) **Phylogenomics reveals multiple losses of nitrogen-fixing root nodule symbiosis** *Science* **361**
- Grele A. *et al.* (2024) **Intra- and interspecific diversity in a tropical plant clade alter herbivory and ecosystem resilience** *eLife* **12**
- Hagan J.G., Vanschoenwinkel B., Gamfeldt L. (2021) **We should not necessarily expect positive relationships between biodiversity and ecosystem functioning in observational field data** *Ecol Lett* **24**:2537–2548
- Hargrave C.W., Hambright K.D., Weider L.J. (2011) **Variation in resource consumption across a gradient of increasing intra- and interspecific richness** *Ecology* **92**:1226–1235
- Hazard C., Kruitbos L., Davidson H., Taylor A.F.S., Johnson D. (2017) **Contrasting effects of intra- and interspecific identity and richness of ectomycorrhizal fungi on host plants, nutrient retention and multifunctionality** *New Phytologist* **213**:852–863
- Hooper D.U., Chapin F.S., Ewel J.J., Hector A., Inchausti P., Lavorel S., et al. (2005) **Effects of biodiversity on ecosystem functioning: A consensus of current knowledge** *Ecol Monogr* **75**:3–35
- Hughes A.R., Inouye B.D., Johnson M.T.J., Underwood N., Vellend M. (2008) **Ecological consequences of genetic diversity: Ecological effects of genetic diversity** *Ecol Lett* **11**:609–623
- Hughes A.R., Stachowicz J.J. (2004) **Genetic diversity enhances the resistance of a seagrass ecosystem to disturbance** *Proc Natl Acad Sci U.S.A* **101**:8998–9002
- Jiang M., He L., Fan B., Wang T., Yang N., Liu Y., et al. (2022) **Intraspecific more than interspecific diversity plays an important role on Inner Mongolia grassland ecosystem functions: A microcosm experiment** . *Sci Total Env* **826**
- Lefcheck J.S. (2016) **PIECEWISESEMIIJ: Piecewise structural equation modelling in R for ecology, evolution, and systematics** *Methods Ecol Evol* **7**:573–579

- Lefcheck J.S., Byrnes J.E.K., Isbell F., Gamfeldt L., Griffin J.N., Eisenhauer N., et al. (2015) **Biodiversity enhances ecosystem multifunctionality across trophic levels and habitats** *Nat Commun* **6**
- Li F., Altermatt F., Yang J., An S., Li A., Zhang X. (2020) **Human activities' fingerprint on multitrophic biodiversity and ecosystem functions across a major river catchment in China** *Global change biology* **26**:6867–6879
- Loreau M., Hector A. (2001) **Partitioning selection and complementarity in biodiversity experiments** *Nature* **412**:72–76
- Loreau M., Jarne P., Martiny J.B.H. (2023) **Opportunities to advance the synthesis of ecology and evolution** *Ecol Lett* **26**
- Macneil C., Dick J.T.A., Elwood R.W. (1997) **THE TROPHIC ECOLOGY OF FRESHWATER GAMMARUS SPP. (CRUSTACEA:AMPHIPODA): PROBLEMS AND PERSPECTIVES CONCERNING THE FUNCTIONAL FEEDING GROUP CONCEPT** . *Biol Rev* **72**:349–364
- Moi D. A., Romero G. Q., Antiqueira P. A., Mormul R. P., Teixeira de Mello F., Bonecker C. C. (2021) **Multitrophic richness enhances ecosystem multifunctionality of tropical shallow lakes** *Funct Ecol* **35**:942–954
- Poisot T., Mouquet N., Gravel D. (2013) **Trophic complementarity drives the biodiversity–ecosystem functioning relationship in food webs** *Ecol Lett* **16**:853–861
- Prieto I., Violle C., Barre P., Durand J.-L., Ghesquiere M., Litrico I. (2015) **Complementary effects of species and genetic diversity on productivity and stability of sown grasslands** *Nat Plants* **1**
- Prunier J.G., Chevalier M., Raffard A., Loot G., Poulet N., Blanchet S. (2023) **Genetic erosion reduces biomass temporal stability in wild fish populations** *Nat Commun* **14**
- Raffard A., Cucherousset J., Montoya J.M., Richard M., Acoca-Pidolle S., Poésy C., et al. (2021) **Intraspecific diversity loss in a predator species alters prey community structure and ecosystem functions** *PLoS Biol* **19**
- Raffard A., Santoul F., Cucherousset J., Blanchet S. (2019) **The community and ecosystem consequences of intraspecific diversity: a meta-analysis: The ecological effects of intraspecific diversity** *Biol Rev* **94**:648–661
- Reusch T.B.H., Ehlers A., Hämmerli A., Worm B. (2005) **Ecosystem recovery after climatic extremes enhanced by genotypic diversity** *Proc Natl Acad Sci U.S.A* **102**:2826–2831
- Schlötterer C., Tobler R., Kofler R., Nolte V. (2014) **Sequencing pools of individuals — mining genome-wide polymorphism data without big funding** *Nat Rev Genet* **15**:749–763
- Seibold S., Cadotte M. W., MacIvor J. S., Thorn S., Müller J. (2018) **The necessity of multitrophic approaches in community ecology** *Trends Ecol Evol* **33**:754–764
- Shipley B. (2009) **Confirmatory path analysis in a generalized multilevel context** *Ecology* **90**:363–368

Soliveres S., van der Plas F., Manning P., Prati D., Gossner M.M., Renner S.C., et al. (2016) **Biodiversity at multiple trophic levels is needed for ecosystem multifunctionality** *Nature* **536**:456–459

Tang T., Zhang N., Bongers F.J., Staab M., Schuldt A., Fornoff F., et al. (2022) **Tree species and genetic diversity increase productivity via functional diversity and trophic feedbacks** *eLife* **11**

Van Der Plas F. (2019) **Biodiversity and ecosystem functioning in naturally assembled communities** *Biological Reviews* **94**:1220–1245

Wan N.-F., Fu L., Dainese M., Hu Y.-Q., Pødenphant Kiær L., Isbell F., et al. (2022) **Plant genetic diversity affects multiple trophic levels and trophic interactions** *Nat Commun* **13**

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Reviewer #1 (Public review):**Summary:**

This work used a comprehensive dataset to compare the effects of species diversity and genetic diversity within each trophic level and across three trophic levels. The results stated that species diversity had negative effects on ecosystem functions, while genetic diversity had positive effects. Additionally, these effects were observed only within each trophic level and not across the three trophic levels studied. Although the effects of biodiversity, especially genetic diversity across multi-trophic levels, have been shown to be important, there are still very few empirical studies on this topic due to the complex relationships and difficulty in obtaining data. This study collected an excellent dataset to address this question, enhancing our understanding of genetic diversity effects in aquatic ecosystems.

Strengths:

The study collected an extensive dataset that includes species diversity of primary producers (riparian trees), primary consumers (macroinvertebrate shredders), and secondary consumers (fish). It also includes genetic diversity of the dominant species in each trophic level, biomass production, decomposition rates, and environmental data. The writing is logical and easy to follow.

Weaknesses:

The two main conclusions-(1) species diversity had negative effects on ecosystem functions, while genetic diversity had positive effects, and (2) these effects were observed only within

each trophic level, not across the three levels—are overly generalized. Analysis of the raw data shows that species and genetic diversity have different effects depending on the ecosystem function. For example, neither affected invertebrate biomass, but species diversity positively influenced fish biomass, while genetic diversity had no effect. Furthermore, Table S2 reveals that only four effect sizes were significant ($P < 0.05$): one positive genetic effect, one negative genetic effect, and two negative species effects, with two effects within a trophic level and two across trophic levels. Additionally, using a $P < 0.2$ threshold to omit lines in the SEMs is uncommon and was not adequately justified. A more cautious interpretation of the results, with acknowledgment of the variability observed in the raw data, would strengthen the manuscript.

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Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

This work used a comprehensive dataset to compare the effects of species diversity and genetic diversity within each trophic level and across three trophic levels. The results showed that species diversity had negative effects on ecosystem functions, while genetic diversity had positive effects. These effects were observed only within each trophic level and not across the three trophic levels studied. Although the effects of biodiversity, especially genetic diversity across multi-trophic levels, have been shown to be important, there are still very few empirical studies on this topic due to the complex relationships and difficulty in obtaining data. This study collected an excellent dataset to address this question, enhancing our understanding of genetic diversity effects in aquatic ecosystems.

Strengths:

The study collected an extensive dataset that includes species diversity of primary producers (riparian trees), primary consumers (macroinvertebrate shredders), and secondary consumers (fish). It also includes the genetic diversity of the dominant species at each trophic level, biomass production, decomposition rates, and environmental data.

The conclusions of this paper are mostly well supported by the data and the writing is logical and easy to follow.

Weaknesses:

(1) While the dataset is impressive, the authors conducted analyses more akin to a "meta-analysis," leaving out important basic information about the raw data in the manuscript. Given the complexity of the relationships between different trophic levels and ecosystem functions, it would be beneficial for the authors to show the results of each SEM (structural equation model).

We understand the point raised by the reviewer. We now provide individual SEMs (Figure 3), although we limit causal relationships to those for which the p-value was below 0.2 for the sake of graphical clarity. We also provide the percentage of explained variance for each ecosystem function. We detail the graph in the Results section (see l. 317-328) and discuss them (see l. 387-398). Note that we do not detail each function separately as this would (in our

opinion) result in a long descriptive paragraph from which it might be difficult to get some key information. Rather, we summarize the percentage of explained variance for each function and discuss the strength of environmental vs biodiversity effects for some examples. In the Discussion, we explain why environmental effects (on functions and biodiversity) are relatively weak. We mainly attribute this to the sampling scheme that follows an East-West gradient (weak altitudinal range) rather than an upstream-downstream gradient as it is traditionally done in rivers. The reasoning behind this sampling scheme is explained in our companion paper (Fargeot et al. Oikos 2023) to which we now refer more explicitly in the MS. Briefly, using an upstream-downstream gradient would have certainly push up the effects of the environment, but this would have made extremely complex the inference of biodiversity effects due to strong collinearity among environmental and biodiversity parameters.

(2) The main results presented in the manuscript are derived from a "metadata" analysis of effect sizes. However, the methods used to obtain these effect sizes are not sufficiently clarified. By analyzing the effect sizes of species diversity and genetic diversity on these ecosystem functions, the results showed that species diversity had negative effects, while genetic diversity had positive effects on ecosystem functions. The negative effects of species diversity contradict many studies conducted in biodiversity experiments. The authors argue that their study is more relevant because it is based on a natural system, which is closer to reality, but they also acknowledge that natural systems make it harder to detect underlying mechanisms. Providing more results based on the raw data and offering more explanations of the possible mechanisms in the introduction and discussion might help readers understand why and in what context species diversity could have negative effects.

(We now provide more details. However, we are unfortunately not sure that this helped reaching some stronger explanation regarding underlying mechanisms. To be frank, we did not succeed in improving mechanistic inferences based on the outputs of the SEM models. We explored visually some additional relationships (e.g. relationships between the biomass of the focal species and that of other species in the assemblage) that we now discuss a bit more, but again, this did not really help in better understanding processes. We realize this is a limitation of our study and that this can be frustrating for readers. Nonetheless, as said in the Discussion, field-based study must be taken for what they are; observational studies forming the basis for future mechanistic studies. Although we failed to explain mechanisms, we still think that we provide important field-base evidence for the importance of biodiversity (as a whole) for ecosystem functions.

1. Environmental variation was included in the analyses to test if the environment would modulate the effects of biodiversity on ecosystem functions. However, the main results and conclusions did not sufficiently address this aspect.

This is now addressed, see our response to your first comment. We now explain (result section) and discuss environmental effects. As explained in the MS, environmental effects are similar in strength to those of biodiversity and are not that high, which is partly explained by the sampling scheme (see Fargeot et al. 2023). This is a choice we've made at the onset of the experiment, as we wanted to focus on biodiversity effects and avoid strong collinearity as it is generally the case in rivers (which impedes any proper and strong statistical inferences).

Reviewer #2 (Public review):

Summary:

Fargeot et al. investigated the relative importance of genetic and species diversity on ecosystem function and examined whether this relationship varies within or between

trophic-level responses. To do so, they conducted a well-designed field survey measuring species diversity at 3 trophic levels (primary producers [trees], primary consumers [macroinvertebrate shredders], and secondary consumers [fishes]), genetic diversity in a dominant species within each of these 3 trophic levels and 7 ecosystem functions across 52 riverine sites in southern France. They show that the effect of genetic and species diversity on ecosystem functions are similar in magnitude, but when examining within-trophic level responses, operate in different directions: genetic diversity having a positive effect and species diversity a negative one. This data adds to growing evidence from manipulated experiments that both species and genetic diversity can impact ecosystem function and builds upon this by showing these effects can be observed in nature.

Strengths:

The study design has resulted in a robust dataset to ask questions about the relative importance of genetic and species diversity of ecosystem function across and within trophic levels.

Overall, their data supports their conclusions - at least within the system that they are studying - but as mentioned below, it is unclear from this study how general these conclusions would be.

Weaknesses:

(4) While a robust dataset, the authors only show the data output from the SEM (i.e., effect size for each individual diversity type per trophic level (6) on each ecosystem function (7)), instead of showing much of the individual data. Although the summary SEM results are interesting and informative, I find that a weakness of this approach is that it is unclear how environmental factors (which were included but not discussed in the results) nor levels of diversity were correlated across sites. As species and genetic diversity are often correlated but also can have reciprocal feedbacks on each other (e.g., Vellend 2005), there may be constraints that underpin why the authors observed positive effects of one type of diversity (genetic) when negative effects of the other (species). It may have also been informative to run SEM with links between levels of diversity. By focusing only on the summary of SEM data, the authors may be reducing the strength of their field dataset and ability to draw inferences from multiple questions and understand specific study-system responses.

We have addressed this remark and we ask the reviewers and the readers to refer to our response to comment 1 from reviewer 1. Regarding co-variation among biodiversity estimates (SGDCs according to Vellend's framework), we have addressed these issues in a companion paper that we now cite and expand further in the MS (Fargeot et al. *Oikos*, 2023). Given the size of the dataset and its complexity (and associated analyses), we have decided to focus on patterns of species and genetic biodiversity in a first paper (*Oikos* paper) and then on the link between biodiversity and functions (this paper). As it can be read in the *Oikos*'s paper, there are no co-variation in term of biodiversity estimates; species diversity is not correlated to genetic diversity, and within facet, there are not co-variation among species. In addition, environmental predictors are highly estimate-specific (i.e. environmental predictors sustaining species and genetic estimates are idiosyncratic). As a result (see the new Figure 3), environmental effects are relatively weak (the same intensity that those of biodiversity) and collinearity among parameters is relatively weak. The second point is important, as this permit to better infer parameters from models, and this allows to discuss direct relationships (as observed in Figure 3, indirect environmental effects are relatively rare). We provide in the Discussion a bit more explanation about the absence of co-variation among biodiversity estimates (see l. 433-440).

(5) My understanding of SEM is it gives outputs of the strength/significance of each pathway/relationship and if so, it isn't clear why this wasn't used and instead, confidence intervals of Z scores to determine which individual BEFs were significant. In addition, an inclusion of the 7 SEM pathway outputs would have been useful to include in an appendix.

We now provide p-values (Table S2) and the seven models (Figure 3).

(6) I don't fully agree with the authors calling this a meta-analysis as it is this a single study of multiple sites within a single region and a specific time point, and not a collection of multiple studies or ecosystems conducted by multiple authors. Moreso, the authors are using meta-analysis summary metrics to evaluate their data. The authors tend to focus on these patterns as general trends, but as the data is all from this riverine system this study could have benefited from focusing on what was going on in this system to underpin these patterns. I'd argue more data is needed to know whether across sites and ecosystems, species diversity and genetic diversity have opposite effects on ecosystem function within trophic levels.

We agree. "Meta-regression" would perhaps be more adequate than "meta-analyses". We changed the formulation.

Reviewer #3 (Public review):

The manuscript by Fargeot and colleagues assesses the relative effects of species and genetic diversity on ecosystem functioning. This study is very well written and examines the interesting question of whether within-species or among-species diversity correlates with ecosystem functioning, and whether these effects are consistent across trophic levels. The main findings are that genetic diversity appears to have a stronger positive effect on function than species diversity (which appears negative). These results are interesting and have value.

However, I do have some concerns that could influence the interpretation.

(7) Scale: the different measures of diversity and function for the different trophic levels are measured over very different spatial scales, for example, trees along 200 m transects and 15 cm traps. It is not clear whether trees 200 m away are having an effect on small-scale function.

Trees identification and invertebrate (and fish) sampling are done on the same scale. Trees are spread along the river so that their leaves fall directly in the river. Traps have been installed all along the same transect in various micro-habitats. Diversity have been measured at the exact same scale for all organisms. We have modified the MS to make this clear.

(8) Size of diversity gradients: More information is needed on the actual diversity gradients. One of the issues with surveys of natural systems is that they are of species that have already gone through selection filters from a regional pool, and theoretically, if the environments are similar, you should get similar sets of species, without monocultures. So, if the species diversity gradients range from say, 6 to 8 species, but genetic diversity gradients span an order of magnitude more, you can explain much more variance with genetic diversity. Related to this, species diversity effects on function are often asymptotic at high diversity and so if you are only sampling at the high diversity range, we should expect a strong effect.

Fish species number varies from 1 to 11, invertebrate family number varies from 15 to 42 and the tree species number varies from 7 to 20 (see Fargeot *et al.* 2023 for details). We have added this information in the M&M. The gradients are hence relatively large and do not cover a restricted set of values. There is a variance in species number among sites, even if sites are collected along a relatively weak altitudinal gradient. This is obviously complex to compare to SNP (genomic) diversity. Genetic and species effects are similar in effect sizes (percentage of explained variance), so it does not seem we have biased one of the two gradients of biodiversity.

(9) Ecosystem functions: The functions are largely biomass estimates (expect decomposition), and I fail to see how the biomass of a single species can be construed as an ecosystem function. Aren't you just estimating a selection effect in this case?

The biomass estimated for a certain area represents an estimate of productivity, whatever the number of species being considered. Obviously, productivity of a species can be due to environmental constraints; the biomass is expected to be lower at the niche margin (selection effect). But if these environmental effects are taken into account (which is the case in the SEMs), then the residual variation can be explained by biodiversity effects. We provide an explanation (l. 217-219).

(10) Note that the article claims to be one of the only studies to look at function across trophic levels, but there are several others out there, for example:

Thanks, we now cite some of these studies (Li *et al.* 2020, Moi *et al.* 2021, Seibold *et al.* 2018).

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Introduction:

The introduction of the manuscript is generally well-structured, and the scientific questions are clearly presented. However, in each paragraph where specific aspects are introduced, the authors do not focus sufficiently on the given points. The current introduction discusses the weaknesses of previous studies extensively but lacks detailed explanations of mechanisms and a clear anticipation of this study's contributions.

For example:

L72-77: The authors mention that "genetic diversity may functionally compensate for a species loss," but this point is not highly relevant to the main analyses of this study, which focus on comparing the relative effects of species diversity and genetic diversity.

Yes true, we understand the point made by the reviewers. We deleted this part of the sentence.

L87-95: As previously noted, "whether environmental variation decreases or enhances the relative influence of genetic and species diversity on ecosystem functions" was not addressed in this study. Additionally, the last sentence seems unnecessary here, as it does not relate to "environmental variation." The phrase "generate insightful knowledge for future mechanistic models" is vague. It would be helpful to specify what kind of knowledge and what types of future mechanistic models are being referred to.

We modified these two sentences. We now posit the prediction that what has been observed under controlled conditions (that genetic and species have effects of similar magnitude)

might not be the norm under fluctuating environments (because it has been shown that environmental variation modulates the strength of interspecific BEFS and create huge variance).

L96-116: The use of "for instance" three times in this paragraph makes the structure seem scattered, as only examples are provided. Improving the transition words can help the text focus better on the main point.

We have modified some parts of this section to better reflect predictions

L115-116: Again, it would be beneficial to specify what kind of insightful information can be provided.

We have modified this sentence by making more explicit some of the information that may be gained.

L117-134: Stating clear expectations can help the introduction focus on the mechanisms and assist readers in following the results.

We now provide some predictions. We were reluctant to make predictions in the first version of the MS as we have the feeling that predictions can go on very different direction depending on how we set the scene. We therefore stick to predictions that we think are the most logical (the simplest ones). This illustrates the lack of theoretical papers on these issues.

Methods:

L287-293: The method for estimating the standard effect size is unclear. I assume it was derived from the SEM models? This needs further clarification.

Yes, it is derived from the standardized estimate from each pSEM. This is now explained in the MS.

Results:

As mentioned in the public review, it is very important to show the results of analyzing raw data.

Done, see Figure 3 and Results section.

Table 1: The font and format of the PCA table are different from other tables and appear vague, resembling a picture rather than a table.

Changed.

Table 2 (and supplementary table): "D.f." is not explained in the table legend. Is 1 the numerator df and 30 the denominator df? Is the denominator the residual? Additionally, the table legend mentions "magnitude and direction." ANOVA only tests if the biodiversity effects are significantly different between species or genetic diversity, but not the magnitude. For example, -0.5 and 0.5 are very different, but their effect magnitudes are the same.

This is a mistake; sorry the format of the Table was from a previous version of the MS in which we used linear models rather than linear mixed models (both lead to the same results). The ANOVA used to test the significance of fixed terms in linear mixed model are based on Wald chi-square tests, and it should have been read "Chi-value" rather than "F-value" in both

tables and the only degree of freedom in this test is the one at the numerator. This has been changed. We have changed the caption of the Table (“ANOVA table for the linear mixed model testing whether the relationships between biodiversity and ecosystem functions measured in a riverine trophic chain differ between the biodiversity facets (species or genetic diversity) and the types of BEF (*within- or between-trophic levels*)”)

Minor:

There should always be a space between a number and a unit. In the manuscript, spaces are inconsistently used between numbers and units.

Corrected

Reviewer #2 (Recommendations for the authors):

(1) In the introduction, the authors could focus more and build out what they predicted/hypothesized as well as what has been found in the manipulated experiments that examined the role of species and genetic diversity. That would enhance the background information for a more general audience, and highlight expected results and why.

We modified the Introduction according to comments made by reviewer 1 and clarified the predictions as best as we can.

(2) Similarly, the discussion is fairly big picture, but this dataset focused exclusively on this 3-trophic interaction in a riverine system. It could be beneficial to dig into the ecology to find out why the opposite effects of species and genetic diversity are seen within trophic levels in this system.

We have added some explanations based on the specific pSEM (see our responses to the public reviews for details). But as said in the responses to the public reviews, even with more detailed models, it is hard to tease apart mechanisms. One important point is that genetic and species diversity do not correlate one to each other (they do not co-vary over space), which means the effect of one facet is independent from the other. However, apart from that, we can't really tell more without more mechanistic approaches. We understand this is frustrating, but this is the nature of field-based data. This does not mean they are useless. On the contrary, they confirm and expand patterns found under controlled conditions (which for ecologists is quite important as nature is our playground), but they are limited in inferences of mechanisms.

(3) It would also be informative if the authors specified what positive and negative Z scores mean. It seems counterintuitive in Figure 3. For example, in the upper left, it's denoted as a larger intraspecific effect - which I'd assume is higher genetic (within species) diversity - but is this not where species diversity effects are higher? In theory this figure could be similar to Figure 1 from Des Roches et al. 2018 - where showing the 1:1 line of where species and genetic diversity effects are similar and then how some are more impacted by SD or GD as that links to the overall question, right?

For example: Figure 3 makes it seem that GD effects are stronger (more positive) for within trophic responses (which is reflected in the text), but in that quadrant, it states that the interspecific effect is larger?

yes, you're true Figure 3 (now Figure 4) is not ideal. We added an explicit explanation for interpreting Z_r in the main text. In addition, we modified the text in the quadrat as this was not correct. Note that it cannot be directly compared to that of DesRoches et al. In DesRoches et al., there is a single effect size (ES) per situation (which is roughly expressed as “ES = effect of species - effect of genotypes”). Here, there are two ES per situation, one for the

species effect, the other for the genetic effect, which makes the biplot more complex (as species and genetic can be similar in magnitude, but opposite in direction, e.g., 0.5 and -0.5). We may have done as DesRoches et al. (“ES = effect of species - effect of genotypes”), but as we don’t have absolute ES (as in DesRoches) the resulting signs of the ES are non sensical...Not easy for us to find a clever solution (or said differently, we were not clever enough to find an easy solution). Nonetheless, we tried another visualization by including “sub-quadrats” into the four main quadrats. We hope this will be clearer

(4) It's unclear why authors included both a simplified linear mixed model with diversity type and biodiversity facet as fixed factors, and then a second linear model that included trophic level (with those other 2 factors and interactions), but only showed results of trophic level from that more complex model. It is unclear why they include two models when the more complex one would have evaluated all aspects of their research question and shown the same patterns.

You’re true, the more complex model evaluates both aspects. Nonetheless, as the hypotheses were strictly separated, we thought it is simpler to associate one model to one hypothesis. We agree that this duplicates information, but we would like to keep the two models to make the text more gradual.

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